Phase	Main hypothesis	What evidence showed
V2-V4	Semantic / detector training	Limited instance PQ (~0.31-0.32)
V7-V8	Native crop inference	V7 advanced to .34; V8 cache mismatch
V9-V11	Ridge / graph / quality rescue	V9 .31, V10 .33; graph benefits not established
V12-V15	Baseline recovery & delta-PQ policies	Some apparent gains against reconstructed wrong controls; frozen .34 p
V16	Fresh semantic/instance model sweep	Best local YOLO11m .3899; public .33
V17-V20	Scale-aware and solar-pretrained detectors	Native-resolution evidence; prior public .36
V22-V25	Quality ranking + selective refinement	V22 .37, V24 .38, V25 .39 public
V26-V29	New recovery and query models	V28 +.000323 / V29 +.000798 local; neither promoted